

Article

Hybrid Boustrophedon and Direction-Biased Region Transitions for Mobile Robot Coverage Path Planning: A Region-Based Multi-Cost Framework

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Abstract

Achieving efficient Coverage Path Planning (CPP) in indoor and semi-structured settings necessitates both organized area segmentation and dependable transitions between coverage zones. This research introduces an improved region-guided CPP framework that incorporates rectangular region expansion, Boustrophedon-based coverage within regions, and an obstacle-aware planner for transitioning between regions. In contrast to conventional methods that depend solely on A*-based routing, the suggested transition module utilizes a multi-weighted cost model that integrates Euclidean distance, obstacle density, and heading changes to create smoother, more context-sensitive links between regions. The approach is assessed on five representative grid maps inspired by the layouts of building corridors and greenhouse-like strip structures. Performance indicators—including intra-region coverage distance, inter-region transition cost, overall path distance, coverage ratio, and computation duration—illustrate the method's efficiency. Experimental findings indicate consistent coverage rates ranging from 96% to 99%, with total computation times between 312 and 844 ms. When compared to traditional global Boustrophedon and spiral scanning methods, the proposed system attains noticeably shorter transition paths and enhanced navigation efficiency, particularly in narrow corridors and cluttered environments. In summary, the framework provides a modular, computationally efficient, and obstacle-aware solution that is well-suited for autonomous mobile robot coverage path planning tasks.

Keywords: Coverage Path Planning; inter-region transition; mobile robots; region obstacle-aware planning; region segmentation



Received: 23 October 2025

Revised: 17 November 2025

Accepted: 20 November 2025

Published: 29 November 2025

Citation: Karakaya, S.; Konyar, M.Z. Hybrid Boustrophedon and Direction-Biased Region Transitions for Mobile Robot Coverage Path Planning: A Region-Based Multi-Cost Framework. *Appl. Sci.* **2025**, *15*, 12666. <https://doi.org/10.3390/app152312666>

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1. Introduction

Recent advancements in mobile robot technology necessitate the ability of these systems to autonomously cover expansive areas in various industrial and service applications. Key applications include industrial cleaning robots [1], agricultural spraying and fertilizing [2], land surveying [3], environmental monitoring [4], search and rescue missions [5], and indoor security solutions [6]. The aim of overarching across all these applications is for the robot to maximize area coverage while completing the specified task with minimal energy, time, and redundancy costs. Coverage Path Planning (CPP) refers to the process of identifying paths that enable a mobile robot to consistently traverse accessible areas and visit every point at least once [7]. In the existing literature, this issue is addressed using a

variety of strategies for known environments (off-line planning) or sensor-based unknown environments (on-line planning) [8]. In particular, in complex environments with structural obstacles, establishing systematic coverage routes is critical for optimizing total coverage efficiency and minimizing energy consumption [9].

CPP methods are generally categorized into two primary approaches: (i) region-based techniques that involve pre-partitioning the area according to its geometry or topology [10], and (ii) global planners that utilize pattern or probability-based methods within the zone [11]. Among these classifications, one widely used technique, boustrophedon decomposition, advocates for dividing the area into sub-regions based on obstacles and ensuring coverage for each region through zigzag scanning in rows [12,13]. Research conducted by Gabriely and Rimon [14] has demonstrated that mobile robots can follow systematic coverage routes that do not overlap in grid-based planning. Acar and Choset [15] have illustrated that topological complexity can be diminished using Morse-based decomposition. Additionally, approaches such as Voronoi-based navigation [16], spiral-based scanning [17], and sensor field modeling [18] have showcased systematic coverage strategies depending on the environment. However, an aspect frequently neglected in existing methods is the impact of inter-region transition paths on total coverage times [19]. While numerous strategies effectively implement intra-region coverage techniques, they often overlook the sequence and expense of inter-region transitions or consider them under static assumptions [20]. In real-world scenarios, the paths that the robot takes between regions significantly influence both energy consumption and overall coverage efficiency [21,22]. This research introduces a comprehensive coverage planning approach to rectify existing gaps in the literature. The suggested system defines expanding rectangular areas on grid-based obstacle maps using a region-growing technique, and generates a boustrophedon coverage path for each area, while the order of transitions between regions is determined by solving the Traveling Salesman Problem (TSP). Additionally, comparative performance evaluations of coverage are conducted utilizing metrics such as coverage efficiency, path length, and operational load [23–25].

In the realm of grid-based coverage and mobile robot path planning, the A* algorithm continues to be a fundamental technique because of its reliable structure and optimal heuristic performance when paired with admissible cost functions. Its extensive use in more recent research [26–30] underscores its role as a standard approach for planning transitions between regions. Therefore, in this research, we utilize A*-based transition logic as a reference point to evaluate the proposed multi-weighted transition planner. This approach not only aligns with the current best practices but also allows for a fair and relevant benchmarking process.

2. Related Work

Coverage Path Planning (CPP) algorithms form the fundamental components in various applications aimed at achieving complete area coverage for mobile robots. In existing literature, these algorithms are generally categorized into two primary groups: (i) Full coverage strategies and (ii) regional planning and optimization methods for transitions.

2.1. Full Coverage Path Planning Approaches

The objective of full coverage methods is for the robot to thoroughly scan all accessible areas at minimal cost. A widely used method known as Boustrophedon decomposition topologically partitions the area and applies a two-way zigzag path to each region [9] proposed dynamic zoning strategies based on this structure to reduce directional changes [26] introduced path strategies that allow the robot to systematically scan by partitioning the workspace into a grid format [27–37] aimed to generate paths in unknown environments

through entropy-based planning. Some research has incorporated physical parameters such as terrain type, multi-robot configurations, energy constraints, and slope details into the coverage planning. However, these techniques often overlook the costs associated with inter-region transitions.

A*-based planning has been widely used in previous CPP research because of its predictable behavior and suitability for grid-based maps. Zhao et al. [36] utilized A* in intricate, energy-limited settings, while Champagne Gareau et al. [38] incorporated it into a branch-and-bound approach to evaluate its performance against their optimal planner. These instances reinforce A*'s position as a standard benchmark for connectivity between regions. According to recent studies, Coverage Path Planning (CPP) aims to ensure complete area coverage with minimal overlap and path redundancy, while optimizing overall execution time and energy consumption in robotic applications [39–41].

2.2. Region-Based Partitioning and Transition Optimization

Region-based CPP strategies facilitate the application of more localized planning strategies by segmenting the area into sub-regions. Acar and Choset [32] executed this decomposition topologically using Morse-based methods. Zelinsky et al. [33] conducted task allocation among robots by dividing the area for decentralized multi-robot coverage arrangements. TSP-based methods condense each area to a central point and seek to minimize transition costs by identifying the shortest path between these points. For instance, Yang et al. [34] enhanced region ranking with TSP-heuristics yet independently planned intra-region coverage paths separate from transition paths.

2.3. Contribution of This Study

The unique contributions of this study to literature are as follows:

1. A systematic and overlap-free creation of rectangular regions that expand maximally on grid-based maps featuring obstacles.
2. The generation of internal paths for full coverage with the least turns, utilizing a boustrophedon pattern CPP approach for each area.
3. Derivation of the shortest total transition path using Multi-Weighted Transition Cost Model.
4. An integrated optimization of both intra-region and inter-region paths using a unified cost function. This integration allows the study to address local and global path planning comprehensively, diverging from traditional CPP methods, and providing a balance between feasibility and efficiency within complex map structures.

3. Problem Definition and Optimization Model

3.1. Problem Statement

Coverage Path Planning (CPP) for autonomous mobile robots seeks to ensure thorough and efficient coverage of restricted and unknown free regions filled with obstacles, whether artificial or natural. This study endeavors to generate cyclically expanding rectangular sub-regions in environments that accommodate obstacle information, advancing beyond the conventional grid-based and often static classical CPP methods. Bidirectional coverage in a boustrophedon format is employed within each region, while the overall coverage trajectory is established by solving the Traveling Salesman Problem (TSP) for transitions between regions. Within this context, the proposed system interactively tackles the following fundamental issues:

- **Region Growing Segmentation:** Development of overlap-free and connected rectangular regions with maximum expansion potential in free areas, considering obstacle information.

- Intra-Region Coverage: Implementation of a Boustrophedon style method for directional and complete coverage within each rectangular area.
- Inter-Region Transition: Reduction of the overall transition cost among regions through TSP-like route optimization centered around each region. The structure introduced adds to the existing literature by offering a distinct coverage strategy capable of decreasing both overall coverage duration and energy expenses in dynamic settings that traditional fixed-grid methods struggle to address.

3.2. Optimization Framework

For a specified map, the goal is to minimize the total coverage route by focusing on these objectives:

- The full free area of the map must be comprehensively covered: $\Omega' = \Omega \setminus \mathcal{O}$.
- Each rectangular section must achieve complete coverage using a Boustrophedon-style path.
- Transitions between different regions need to be optimized to reduce the total distance.

Decision Variables:

- $R = \{r_1, r_2, \dots, r_n\}$: Rectangular coverage areas formed in unblocked sections
- $p_i \in R^2$: Center coordinates for every region
- $T = \{t_1, t_2, \dots, t_n\}$: TSP cycle that determines the order of visits
- $P_i = \{(x_k, y_k)\}_{k=1}^{m_i}$: Boustrophedon coverage path generated within region r_i

Objective Function:

The overall cost function, which integrates coverage within regions and transitions between regions (Equation (1)), employs a cost decomposition approach that has been used in recent optimization studies focused on regions [14,16,34,35]. The aim is to minimize the total cost C_{total} : This cost represents the aggregate length of paths for intra-region coverage as well as inter-region transitions (Equation (1)):

$$C_{\text{total}} = \sum_{i=1}^n C_{\text{CPP}}(P_i) + \sum_{j=1}^{n-1} |p_{T_j} - p_{T_{j+1}}| \quad (1)$$

Here:

- $C_{\text{CPP}}(P_i)$: Path length necessary for the intra-region Boustrophedon coverage
- $|p_{T_j} - p_{T_{j+1}}|$: Euclidean distance between two successive region centers

Constraints:

- Full Coverage: Each unblocked cell must be covered exactly once; overlaps or gaps are prohibited (Equation (2)).

$$\forall (x, y) \in \Omega' \Rightarrow \exists ! r_i : (x, y) \in r_i \wedge (x, y) \in P_i \quad (2)$$

- Transition Compatibility: Each region in the TSP sequence must be visited exactly once (Equation (3)).

$$\forall i \neq j, T_i \neq T_j \quad (3)$$

- Transition Path Validity: Paths for inter-region transitions must not interfere with obstacles. In this study, transition paths are enhanced to navigate around impediments between region segments employing the Multi-Weighted Transition Cost Model.

4. Proposed Methodology

4.1. Methodological Overview

This research introduces a consolidated coverage planning framework that guarantees autonomous robots will systematically and efficiently cover all accessible spaces in grid-based environments cluttered with fixed obstacles. Figure 1 summarizes the proposed method's workflow. Initially, the input obstacle grid map is segmented into rectangular regions (Region Growing). Full coverage is then conducted within each region using a directional boustrophedon pattern (Intra-Region CPP). Smaller and more fragmented regions are merged according to the FSM (Fill-Similarity Measure) ratio to boost structural efficiency. The transition path connecting the centers of the new regions is optimized using the solution to the Traveling Salesman Problem (TSP) (Inter-Region TSP). Ultimately, the intra-region and inter-region paths are integrated into a single framework to create a Comprehensive Global Coverage Path.

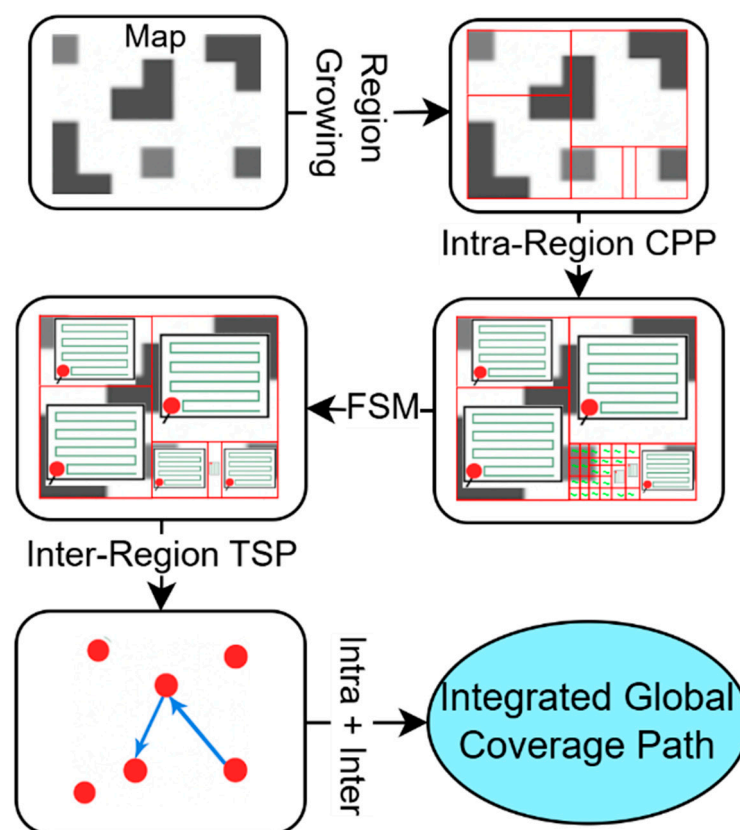


Figure 1. General structure of the proposed coverage planning system.

The introduced method features a multi-stage, modular framework comprising the following components:

- Geometric Obstacle-Aware Region Growing Segmentation: Division of obstacle-free areas into contiguous and uniform rectangular regions optimized for maximum expansion.
- Directional Boustrophedon Based Coverage: Ensuring full coverage with a path generated in a regular zig-zag manner for each rectangular sub-region.
- TSP-Solution for Minimum Transition Route between Regions: Optimizing the transitions between the centers of each region using a Nearest Neighbor-based solution to the Traveling Salesman Problem (TSP) to minimize the overall path length.

Additionally, to more accurately represent the modular dependencies of the proposed framework, directional arrows have been incorporated to show the flow of information

between the Region Growing, Intra-region CPP, and FSM–Inter-region TSP modules. For example, the FSM unit manages both intra- and inter-region transitions utilizing the path outputs from local CPP blocks along with the data from the region center. These feedback loops are essential for producing a cohesive global coverage path and have now been visually highlighted in Figure 1.

The primary objective of this comprehensive structure is to overcome traditional segmentation strategies; that is, to generate regions that are formally compatible and non-overlapping in complex environments where fixed grid or heuristic division methods are insufficient, and to create an optimal global energy-distance coverage plan. In this study, the robot is represented as a point, and map inflation is utilized on all obstacles to represent the true collision boundary according to the robot's physical dimensions.

4.2. Grid-Based Obstacle-Aware Region Growing

In this phase, the preprocessed map's free (obstacle-free) cells are segmented into maximum-sized rectangular coverage regions through regular expansion. The algorithm verifies the following conditions while inspecting the growth of the regions horizontally and vertically from their starting points:

Expansion Condition:

- $\text{map}(i, j) = 0$ and $\text{used}(i, j) = 0$: The target cell must be unoccupied and not utilized previously.
- Expansion proceeds as long as there are gaps in all four directions (north, south, east, and west).

Region Definition:

- Each region is represented by a coordinate box.
- All cells within r_{min} , r_{max} , c_{min} , c_{max} belong to region R_i (Equation (4)).

$$r_{min} \leq i \leq r_{max}, c_{min} \leq j \leq c_{max} \quad (4)$$

It is important to note that this phase, unlike traditional cell decomposition techniques, has an expansion constraint that adjusts to the distribution of obstacles and seeks to minimize the number of regions. Each new region is identified in a manner that avoids overlapping with existing ones by expanding to cover the maximum area.

4.3. Coverage Path Planning (Boustrophedon Scanning) and Inter-Region Path Planning (TSP)

For every rectangular sub-region, the robot's path is established using a row-based zig-zag (Boustrophedon) pattern. The goal is to cover each free cell just once while minimizing the number of changes in direction throughout this process. The path is aligned in a manner that fits the geometry of the region (parallel to its width).

Definition of Waypoints: The coverage path assigned to each region is described as follows (Equation (5)):

$$P_k = \{(x_{k1}, y_{k1}), (x_{k2}, y_{k2}), \dots, (x_{km_k}, y_{km_k})\} \quad (5)$$

Total Intra-Region Path Length The total of the path lengths calculated for all regions is expressed as follows (Equation (6)):

$$L_{intra} = \sum_{k=1}^{N_r} \sum_{l=1}^{m_k-1} |P_k(l+1) - P_k(l)|_2 \quad (6)$$

Here $|\cdot|_2$, denotes the Euclidean distance. The zig-zag scanning pattern can be automatically determined based on both the operating amplitude of the robot and the ratio of latitude to longitude.

Each rectangular coverage area is characterized by its geometric center. The shortest path connecting these centers is determined by finding a solution to the Traveling Salesman Problem (TSP).

Region Center Definition: When a region is labeled as $R_k = (r_{\min}, r_{\max}, c_{\min}, c_{\max})$, the coordinates of its center are defined as follows (Equation (7)):

$$C_k = \left(\frac{c_{\min} + c_{\max}}{2}, \frac{r_{\min} + r_{\max}}{2} \right) \quad (7)$$

Total Cost of Transition: The overall cost of the TSP tour among regional centers is given by (Equation (8)):

$$L_{\text{inter}} = \sum_{i=1}^{N_r-1} |C_{T_i} - C_{T_{i+1}}|_2 \quad (8)$$

In this case, $T = \{T_1, T_2, \dots, T_{N_r}\}$, represents the permutation that dictates the visiting order. This order can be derived using the Nearest Neighbor algorithm or other alternative TSP solutions. This step facilitates the efficient integration of the global path plan in scenarios involving multi-region coverage while significantly influencing overall energy consumption and coverage duration.

4.4. Region Merging via Fill-Similarity Measure (FSM)

Although the rectangular regions generated during the region growing phase are created with a focus on maximum expansion, some regions may become compressed into narrow spaces due to the map's geometry and obstacle layout, resulting in them being relatively small in size. Such small regions lower overall coverage efficiency, increase the frequency of direction changes, and prolong the robot's movement time. As a result, a metric known as the Fill-Similarity Measure (FSM) is proposed to merge regions that are smaller than a specified limit established after the region growing process with their suitable larger adjacent regions. This metric quantitatively assesses the coverage efficiency of the rectangular union between two regions and facilitates merge decisions based on geometric considerations.

FSM Ratio Definition: The FSM ratio for two regions R_i and R_j is defined as follows (Equation (9)):

$$\text{FSM}(R_i, R_j) = \frac{A(R_i) + A(R_j)}{A(R_i \cup R_j)} \quad (9)$$

Here:

- $A(R_k)$: Area of region R_k (measured in pixels or number of cells)
- $R_i \cup R_j$: The least common rectangle enclosing the boundaries of both regions

The closer this ratio is to 1, the more "gapless" and efficient the newly formed region from the merge will be. In practical terms, a threshold value τ is defined for the FSM ratio (for example, $\tau = 0.85$). If the $\text{FSM}(R_i, R_j) \geq \tau$ condition is met, these two regions are considered suitable for merging.

Merge Algorithm:

- **Detection of small regions:** Regions with an area less than $A_{\min}$ are classified as small.
- **Searching for the optimal match:** For each small region, the large neighboring region with the highest FSM ratio is determined.

- Greedy merge: is executed on the first match whose FSM ratio surpasses the defined threshold.
- Reassignment of temporary regions: Small regions that cannot be merged are “shadowed” to the nearest large region.

Figure 2 illustrates the progressive and sequential structure of the region merging algorithm based on the FSM (Fill-Similarity Measure) ratio. Initially identified solely as a large area, R_5 was assessed sequentially alongside the small areas that were geometrically adjacent to it, and the merge was conducted according to their suitability status. First, when evaluating the R_4 area in terms of FSM ratio with R_5 , it remained above the established threshold value ($\theta = 0.85$), yielding an FSM_{45} value of 0.92. This ratio indicates that the common rectangular boundary formed by the combination of the two areas is largely filled. Consequently, R_4 was merged with R_5 to create a new combined area, R_{45} . Following this integration, the subsequent small area, R_3 , was evaluated against R_{45} , once again exceeding the threshold value with FSM_{345} equating to 0.89. The FSM ratio indicates that the merger is effectively efficient, even with a slight reduction in the occupancy rate. In a similar manner, the assessment for the R_2 area resulted in a value of $FSM_{2345} = 0.86$, confirming the merger as suitable. However, the smallest area, R_1 , although physically next to R_{2345} , was calculated to have an FSM ratio of $FSM_{12345} = 0.78$. Since this value fell below the predefined threshold, the R_1 area was excluded from the system, and the merger did not proceed. Thus, an effective, controlled, and limited area expansion strategy was executed by considering not only geometric adjacency but also occupancy rates. The FSM ratios calculated for each merger iteration and the decisions made based on these ratios are compiled in Table 1.

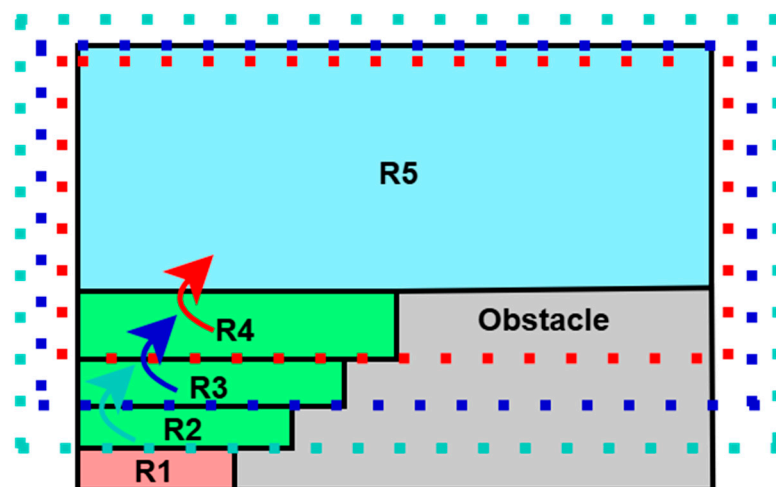


Figure 2. FSM-based chained region merging steps.

Table 1. Step-by-step region merging process and decision status based on FSM ratio.

Stage	Merging Regions	FSM Rate	Decision
FSM_{45}	$R_5 + R_4$	0.92	Merged
FSM_{345}	$R_{45} + R_3$	0.89	Merged
FSM_{2345}	$R_{345} + R_2$	0.86	Merged
FSM_{12345}	$R_{2345} + R_1$	0.78	Excluded

An elevated FSM ratio (approaching 1.0) signifies that the combination of two areas forms a highly effective rectangular space with minimal voids, which is advantageous for decreasing unnecessary directional shifts and enhancing path compactness. In contrast,

a lower FSM ratio indicates that merging could lead to inefficiencies, such as disjointed shapes or ineffective space usage. Therefore, regions yielding FSM values below a certain threshold (e.g., $FSM < 0.85$) are omitted from merging to ensure high coverage quality. Consequently, the FSM metric strikes a balance between geometric compactness and the viability of merging, thereby enhancing overall path planning efficiency.

This approach can be represented as a specific variant of the classical “Set Union” and “Area Optimization” problems. The FSM ratio serves as a ratio-based objective function aiming to maximize the filling efficiency of the rectangular area that will result from the union. Therefore, it prevents the inclusion of unnecessary gaps within the coverage area and overly small areas into the system. This union process substantially enhances the path planning efficiency of the system by operating with fewer but larger areas, thereby reducing the total cost of the integrated objective function. The algorithm is structured for low computational cost, making it suitable for implementation in the MATLAB environment.

4.5. Obstacle-Aware and Directionally Biased Transition Planning

In the standard version of our framework, transitions between regions are determined using a greedy selection approach along with A*-based pathfinding. Although this combined solution provides an efficient and practical method for path construction in organized environments, it has four key limitations when used in maps characterized by a high density of obstacles or restricted spatial arrangements.

4.5.1. Limitations of A*-Based Inter-Region Transition

Although A*-based methods are widely used as a standard in grid-based planning, their implementation during inter-region transitions reveals significant challenges, particularly as the complexity of the environment increases. The limitations discussed here emphasize critical issues concerning spatial reasoning, obstacle detection, and planning methodologies.

- (i) **Dominance of Euclidean Proximity:** In the A*-based method, the main criterion for decision-making is the Euclidean distance between the endpoints of segments. This distance-focused approach overlooks important semantic factors such as risk levels, passage width, and the closeness to obstacles.
- (ii) **Insensitivity to Obstacle Density:** While A* effectively avoids collisions, it fails to differentiate between narrow passageways and wide corridors, nor does it impose penalties on paths that traverse areas with high densities of obstacles. This can lead to transitions that are technically valid but not feasible in practice.
- (iii) **Ignorance of Heading Variation:** The traditional A* planning cost function does not consider the angular discrepancy between the robot's orientation and the direction of the transition. This oversight increases energy expenditure for differential or nonholonomic robots.
- (iv) **Short-Sighted Optimization:** Greedy sequencing of segments aims to minimize local costs without considering the overall optimality of the entire tour. This drawback becomes more pronounced in larger maps, where the effects of inefficient transitions accumulate over time.

4.5.2. Proposed Multi-Weighted Transition Cost Model

To address the shortcomings highlighted in the previous section, it is proposed a more exhaustive transition model. This enhanced approach builds upon the standard A*-based transition planner by incorporating a multi-objective cost function that considers not only geometric distances but also environmental constraints and directional consistency. In this subsection, we will detail the elements of the proposed cost function and its mathematical

representation. To overcome these challenges, we propose a multi-objective cost function defined as (Equation (10)):

$$C_{i \rightarrow j} = \alpha \cdot d_{\text{euclid}}(p_i, p_j) + \beta \cdot \rho_{\text{obs}}(p_i, p_j) + \gamma \cdot \delta_{\text{angle}}(p_i, p_j) \quad (10)$$

where

- $d_{\text{euclid}}(p_i, p_j)$: The Euclidean distance from the exit of region R_i to the entry of region R_j .
- $\rho_{\text{obs}}(p_i, p_j)$: The normalized density of obstacles along the transition route identified by A^* .
- $\delta_{\text{angle}}(p_i, p_j)$: The absolute difference in angle regarding heading direction.
- α, β, γ : Adjustable scalar weights.

The scalar parameters α , β , and γ used in the proposed transition cost function are essential for balancing geometric, environmental, and kinematic aspects. The selection of these parameters is dependent on the specific application. For example, in environments that are cluttered, like greenhouses, it is beneficial to assign a higher value to β to impose penalties for transitions occurring close to obstacles. On the other hand, in spacious corridors or open fields, it may be more advantageous to focus on α to prioritize shorter routes. When dealing with nonholonomic robots, where the cost of turning is significant, it is important to set γ to a higher value to encourage transitions that maintain directional consistency. In our approach, we employed a grid search method across typical scenarios to identify suitable values (e.g., $\alpha = 1.0$, $\beta = 2.0$, $\gamma = 1.5$). This modular structure also facilitates the future integration of learning-based techniques for parameter tuning.

This cost function promotes transitions that are not only more concise but also navigate through less congested spaces while preserving a smooth directional flow. The obstacle density is approximated as (Equation (11)):

$$\rho_{\text{obs}} = \frac{1}{L} \sum_{k=1}^L m(x_k, y_k) \quad (11)$$

where (x_k, y_k) represent the grid coordinates along the A^* path, $m(\cdot) \in \{0, 1\}$ and is the binary occupancy map.

In a similar fashion, the cost associated with heading deviation is articulated as (Equation (12)):

$$\delta_{\text{angle}}(p_i, p_j) = \left| \theta_i - \tan^{-1} \left(\frac{y_j - y_i}{x_j - x_i} \right) \right| \quad (12)$$

where θ_i represents the robot's orientation at the exit location of region R_i .

4.5.3. Incorporation into the Global Planner

Incorporating the proposed cost model into the global planner entails reinterpreting the inter-region transition as a graph traversal problem. Each endpoint of the region segment acts as a node in the graph, and the feasible transitions are represented as weighted edges. The cost model substitutes the traditional Euclidean metric to facilitate more informed transition choices throughout the planning horizon. The subsequent steps outline the integration methodology.

Let $G = (V, E)$ be a directed graph where V represents the endpoints of segments and E denotes the viable transition paths assessed through A^* . Each edge $e_{ij} \in E$ is assigned a weight $C_{i \rightarrow j}$, as previously defined. The planner progressively develops a tour over G by utilizing a modified greedy approach with constrained lookahead:

- Begin with any chosen starting segment R_s .

- At each iteration, assess all possible transitions to unvisited segments by employing $C_{i \rightarrow j}$.
- Choose the segment that has the lowest weighted cost.
- Continue this process until every segment has been visited.

This approach produces a route that effectively balances travel efficiency, smooth maneuverability, and environmental consciousness.

Beyond its theoretical foundation, the proposed transition model delivers several practical advantages that improve the planner's effectiveness in real-world robotic applications. These benefits are summarized as follows:

- Enhanced alignment with nonholonomic constraints,
- Flexible for cluttered or semi-structured settings,
- Works well with real-time grid-based systems,
- Scalable to medium-scale maps (approximately 50×50 grids) with planning times under 1000 ms.

This model more effectively connects local path generation with global path optimization compared to planners that rely solely on A* or Euclidean transitions. Validation through experiments is presented in the Results Section.

The overall structure of this system is illustrated in Figure 3. In the figure's notation, each R_k denotes a region formed in the coverage plan. Each region is designed to have two endpoints (p_1, p_2). The term $L_{\text{Int}k}$ represents the length of the Boustrophedon-style coverage (CPP) path implemented within the R_k region; L_k signifies the cost of Multi-Weighted Transition based connection path employed to shift to the next region. Together, these two elements make up the total connection cost for the respective step (Equation (13)):

$$L_{\text{total}k} = L_{\text{Int}k} + L_k \quad (13)$$

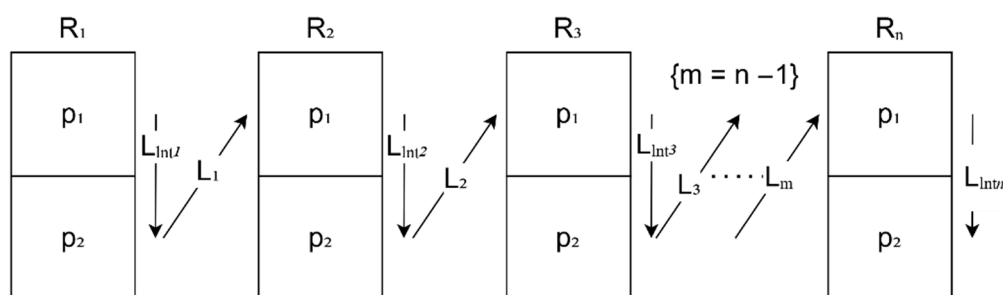


Figure 3. Connection model based on intra-coverage and switching costs.

Here, k signifies the sequence order of connections in the segment. Throughout each transition, a greedy strategy is utilized, favoring the connection with the minimum $L_{\text{total}k}$ value.

Optimizing both intra-region coverage paths (intra-region CPP) and inter-region transition paths (inter-region TSP) is vital for the overall performance of the system. Consequently, the final objective function of this study is presented in a weighted form based on the total costs of both components (Equation (14)):

$$\min(L_{\text{intra}} + \lambda \cdot L_{\text{inter}}) \quad (14)$$

where

- L_{intra} : Sum of boustrophedon path lengths across all rectangular areas.
- L_{inter} : Total length of the transit route between centers of regions.
- $\lambda \in R^+$: User-definable scale factor that modifies the weight of the transition cost.

This formulation allows for comprehensive coverage planning within the area and strategic transitions between regions to be addressed within a single integrated framework. It provides solution adaptability, especially in maps with multiple regions, and system dynamics can be adjusted using parametric modifications. This method is designed for application in the MATLAB environment to optimize both coverage path efficiency and the total distance for inter-regional transition routes. The entire mathematical framework is supported by validation scenarios on complex multi-regional maps. The subsequent pseudocode presents the proposed algorithm for planning transitions between regions, merging spatial, environmental, and directional factors into a cohesive decision-making process (Algorithm 1).

Algorithm 1. Obstacle-Aware Inter-Region Transition Planner.

Input:
 $R = \{R_1, R_2, \dots, R_n\}$ //Set of rectangular regions
 p_i, θ_i //Exit point and heading of each region
 map_data //Binary occupancy grid
 α, β, γ //Weight parameters
Output:
 P_global //Ordered list of region visits
1: Initialize $S \leftarrow \{1, \dots, n\}$, $P_global \leftarrow []$
2: $current \leftarrow SelectInitialRegion(S)$
3: Remove current from S
4: while $S \neq \emptyset$ do
5: $min_cost \leftarrow \infty$
6: for each $j \in S$ do
7: $path_astar \leftarrow AStar(p_current, p_j, map_data)$
8: $d \leftarrow EuclideanDistance(p_current, p_j)$
9: $\rho \leftarrow MeanOccupancy(path_astar)$
10: $\delta \leftarrow |\theta_current - atan2(\Delta y, \Delta x)|$
11: $C \leftarrow \alpha \cdot d + \beta \cdot \rho + \gamma \cdot \delta$
12: if $C < min_cost$ then
13: $min_cost \leftarrow C$
14: $next \leftarrow j$
15: $best_path \leftarrow path_astar$
16: end if
17: end for
18: Append next to P_global
19: $current \leftarrow next$
20: Remove next from S
21: end while
22: Return P_global

In each iteration, the planner assesses all possible transitions from the current region to unvisited options through a multi-weighted cost function. While A* is utilized as a subroutine to identify collision-free routes between segment endpoints, the choice of transition is directed by a specialized cost model that incorporates Euclidean distance, obstacle density, and heading deviation. This approach facilitates a more adaptable and realistic strategy for traversing regions, particularly in environments that are cluttered or semi-structured. The result is a globally ranked list of region visits that reduces the overall transition cost while maintaining local coverage completeness.

5. Results

To assess the effectiveness of the proposed method that combines region growing, Boustrophedon coverage, and inter-regional TSP based path planning, the following performance metrics are considered:

- Number of Regions (N_r): Total number of non-overlapping rectangular sub-regions generated from the Region growing algorithm.
- Intra-Region Coverage Path Length (L_α): The cumulative length of coverage paths produced by the Boustrophedon scanning method in each rectangular region.
- Inter-Region Transition Cost (L_{int}): The total length of transition paths determined by the Traveling Salesman Problem (TSP) approach, solved via the nearest neighbor method, through the center point of each region.
- Total Path Cost (L_{total}): The overall cost of coverage and transition paths (Equation (15)):

$$L_{total} = \sum L_\alpha + L_{int} \quad (15)$$

- The proportion of the total area covered by the robot compared to the accessible (obstacle-free) zones on the map.
- Running Time (s): The duration required to execute the entire algorithm (including segmentation, coverage, and transition planning).

5.1. Test Maps and Setup

Figure 4a presents Map-1, a representative testing environment that has been digitally reconstructed from the actual floor plan of the Faculty of Engineering where this research took place. The map encompasses a section measuring 30 m $\times$ 30 m, featuring large classrooms, corridor intersections, clusters of offices, and open-access spaces. Each area is divided using axis-aligned rectangular sections that correspond to the building's structural design, ensuring that the orientation of regions aligns with the prevailing architectural directions.

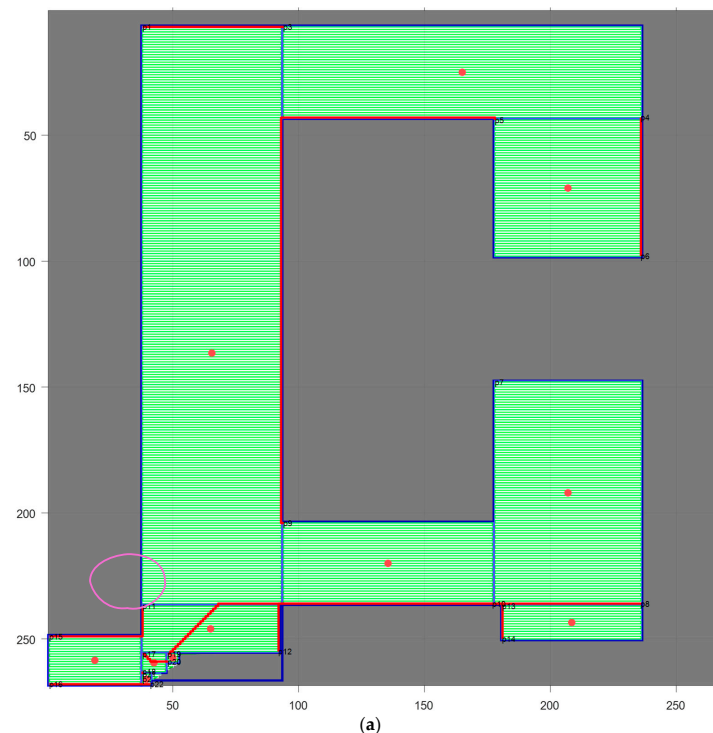


Figure 4. Cont.

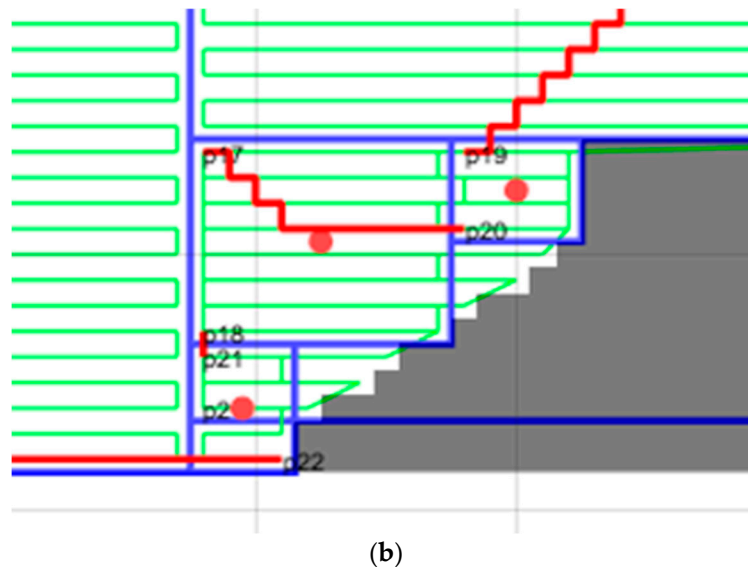


Figure 4. (a). Structural Building Layout (Map-1). (b). Irregular Geometry Evaluation.

In this setting, the majority of transitions happen along horizontal and vertical hallways that are largely unobstructed, enabling the proposed planner to exhibit its greatest efficiency in environments characterized by low clutter and structured layouts. The segmentation of regions is distinctly aligned with spatial boundaries, preventing any overlap or fragmentation.

The inset zoomed in at the bottom of the figure emphasizes an important aspect of the algorithm: the process of merging regions. In locations where fragmentation is excessive—such as tightly knit stair-step-like sub-regions—neighboring regions that possess compatible coverage paths are selectively merged to create larger, more cohesive areas. This action, illustrated by the merged green regions and simplified inter-paths (P_{17} to P_{22}), decreases the number of necessary transitions while ensuring complete coverage. The merging process is steered by factors like adjacency, alignment of coverage, and continuity of obstacles, effectively highlighting the planner's flexibility in transforming overly segmented structures into optimized macro-regions. The visual symbols used in Figures 5–9 are explained in Table 2.

In Figure 4b, there are numerous irregular edges, particularly in the central and lower-left areas, where free-space corridors do not align perfectly with horizontal or vertical lines. Although the region growing process is based on an axis-aligned framework, the FSM-based techniques for region merging and post-processing enable the method to accommodate these atypical shapes by: Allowing for partial fills in regions bounded by diagonal lines, Preventing excessive segmentation of curved or angled areas, and Effectively sustaining coverage with minimal changes in direction.

Figure 5 depicts Map-2, a grid-based environment that has a semi-structured design intended to replicate repetitive indoor configurations such as greenhouses, storage facilities, or industrial shelving systems. The workspace features parallel longitudinal sections aligned along the horizontal axis, interspersed with sparse, strategically placed obstacles that partially influence the planner's inter-region transition behavior without creating complete blockages.

Each rectangular section is similar in size, allowing for uniform intra-region Boustrophedon coverage. Nonetheless, the placement of obstacles introduces asymmetries in transitions between regions, particularly within the lower region cluster, where changes in direction and merging patterns become increasingly important. The segment layout in the

central vertical corridor reveals symmetrical constraints that mimic aisle-based movement patterns typical in agricultural row coverage or palletized storage designs.

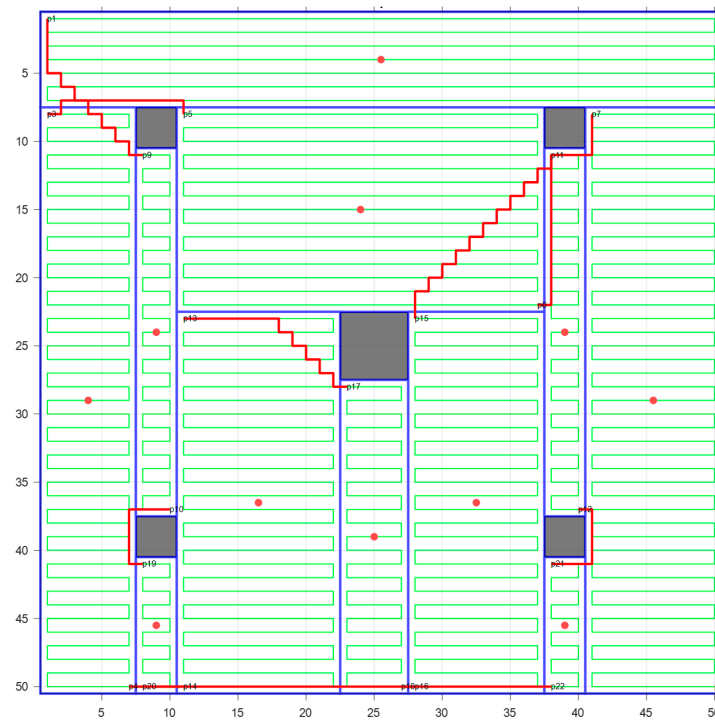


Figure 5. Dense Free Grid Structure (Map-2).

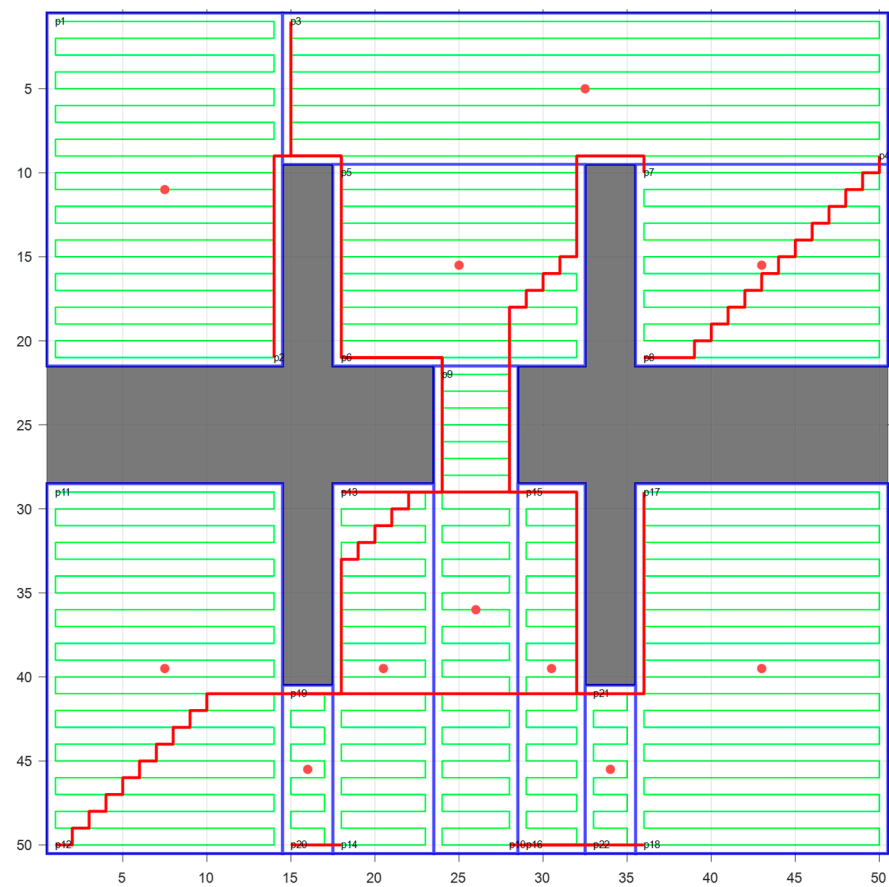


Figure 6. Symmetrical Gate Structure (Map-3).

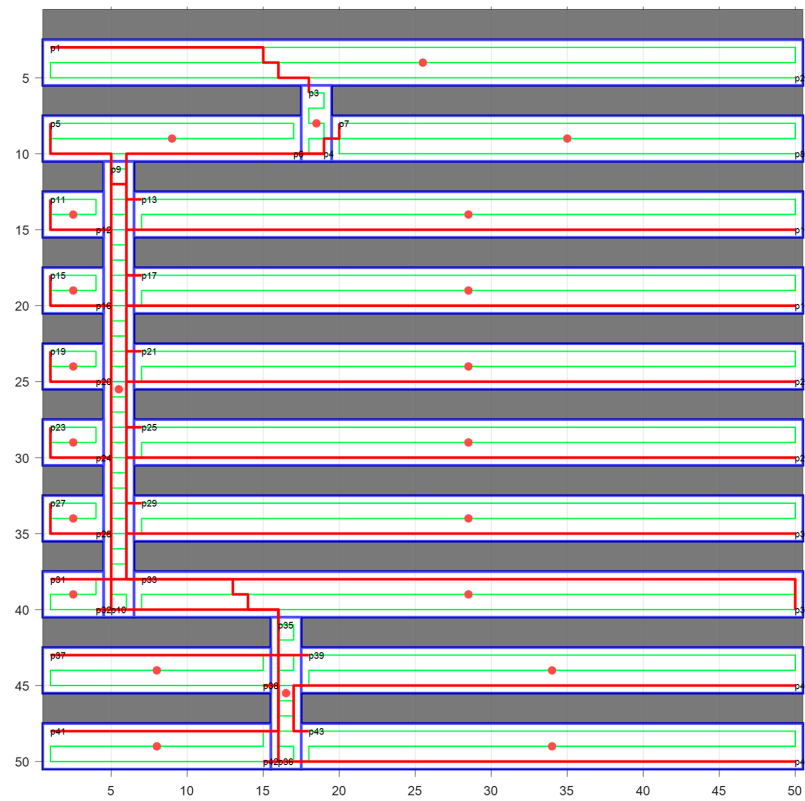


Figure 7. Agriculture/Greenhouse-like Strip Structure (Map-4).

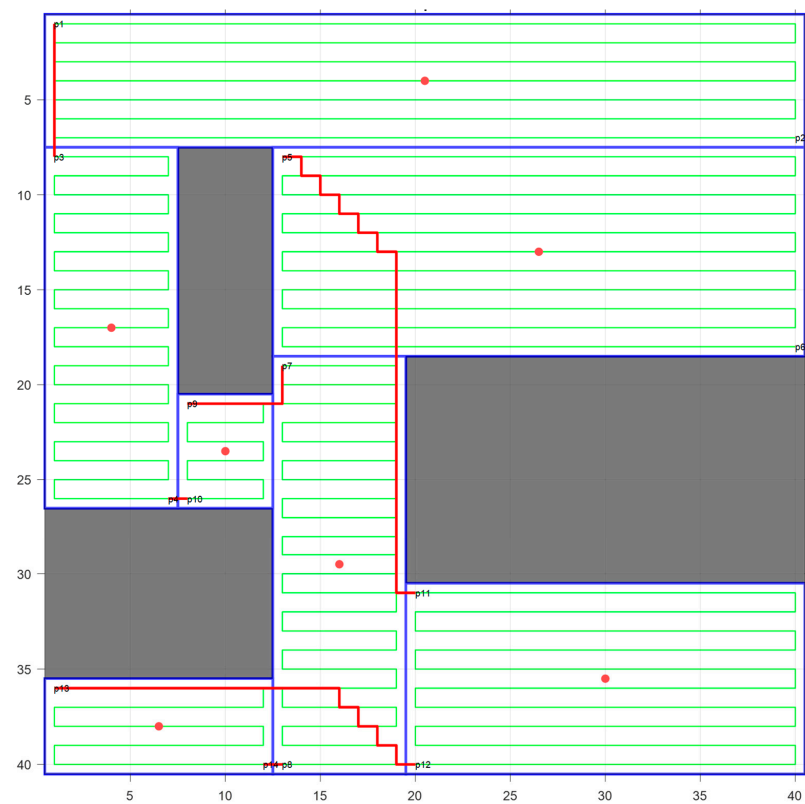


Figure 8. Semi-structured Mixed Geometry (Map-5).

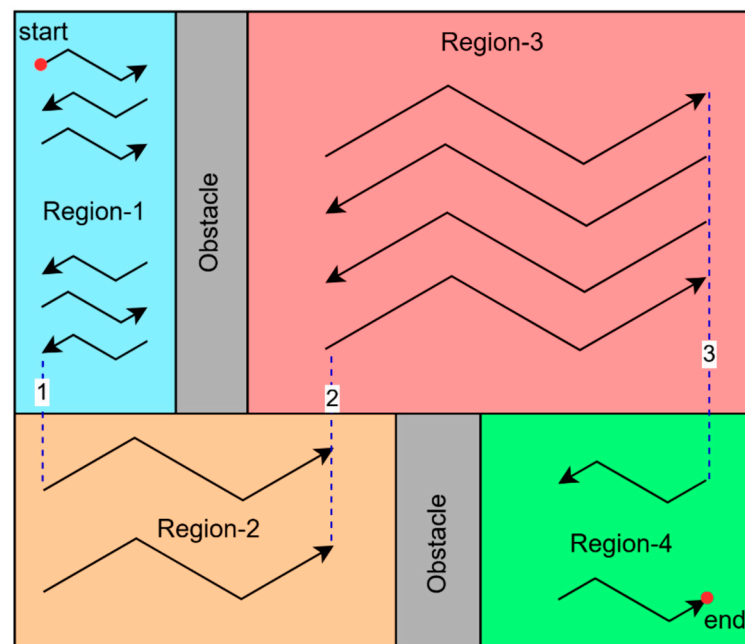


Figure 9. Zigzag coverage paths and TSP-based inter-region passages.

Table 2. Symbol legend table.

Symbol	Definition	
	Light Green Line	Coverage path (Boustrophedon sweep)
	Red Line	Inter-region path
	Red Dot over Green-Hatched Box	Region center
	Blue Line	Region boundary

This map was chosen to assess the planner's ability to take advantage of structural repetition, reduce unnecessary turns, and ensure smooth navigation in environments characterized by high coverage regularity, while still requiring obstacle-aware transition planning. The algorithm's adaptable response to this semi-uniform setup illustrates both its reliability and its ability to react to slight architectural irregularities.

Map-3 (Figure 6) represents a symmetrically partitioned gated environment, featuring a prominent central obstacle that imposes architectural constraints by dividing the map into four distinct zones. The cross-shaped obstruction serves as a choke point, enforcing narrow and centralized passageways that challenge both coverage consistency and transition feasibility.

This layout serves as a critical benchmark for evaluating the proposed Obstacle-Aware Inter-Region Transition Planner, which must navigate limited-access gates while preserving optimal segment visitation. Unlike uniform or semi-structured maps, here the planner is required to reason about passage availability, directional continuity, and symmetrical alignment, all of which interact nontrivially with the cost function's angular deviation and obstacle density terms.

Notably, segment orientations are adaptively aligned to the map's architectural symmetry, and the planner successfully constructs transitions that minimize costly detours through narrow gates. The resulting trajectory highlights the framework's capability to maintain coverage efficiency even under severe spatial constraints.

This scenario approximates environments such as automated greenhouses with zonal gates, controlled-access warehouses, or modular laboratory floors, where efficient and adaptive inter-region routing is paramount.

Map-4 (Figure 7) models an agricultural or greenhouse-like setting featuring a series of elongated, narrow segments, each with small stationary obstacles placed at regular intervals. These obstacles create minor interruptions within the otherwise straight coverage strips, mimicking real-world elements such as planting rows, irrigation systems, or pathways.

The map provides a demanding situation for planning transitions between regions, requiring the robot to execute brief, frequent, and accurately directed movements while steering clear of overlap and dead ends. The suggested Obstacle-Aware Inter-Region Transition Planner shows great proficiency in handling these frequent navigation challenges by utilizing its angular bias and penalties for obstacle density. Importantly, the planner maintains directionality along the horizontal axis, minimizing the need for frequent heading changes and ensuring smooth movement.

This setup is particularly applicable to precision agriculture, automated greenhouse operations, and tasks involving inspection in narrow lanes, where traversal limitations are stringent and collision tolerance is low. The system's capacity to create locally optimized, high-frequency transitions in tight spaces further underscores its effectiveness in confined and repetitive environments.

Map-5 (Figure 8) illustrates a semi-structured design with a combination of geometric patterns, where rectangular coverage zones are positioned both horizontally and vertically around a prominent central obstacle. This arrangement is especially pertinent for hybrid indoor environments, like multifunctional office spaces or adaptable warehouse areas, where fixed partitions and irregular spatial uses coexist.

The large rectangular obstacle located at the center acts as a navigational bottleneck, requiring intelligent routing between regions to ensure continuous overall coverage. The algorithm effectively tackles this issue through its Obstacle-Aware Inter-Region Transition Planner, which assesses visibility corridors and detour costs dynamically during the path generation process. The transitions in this map are primarily vertical and need to be calculated with high accuracy due to frequent obstructions and narrower corridor widths.

Additionally, the irregular shapes and orientations of segments challenge the algorithm's adaptive segmentation strategy, demonstrating its strength in settings that stray from standard orthogonal or uniform grid layouts. The framework successfully illustrates context-sensitive transition generation while maintaining total coverage and execution time, highlighting its practical effectiveness for real-world semi-structured designs.

Table 3 presents a quantitative evaluation of the proposed algorithm's performance across five different map configurations. It is clear from the metrics that Map-1 exhibits the greatest intra-segment coverage path length $L_\alpha = 797$, mainly due to its large and sparse areas, while also experiencing the longest inter-region transitions $L_{int} = 3333$, leading to the highest total path cost $L_{total} = 34,130$ and a processing time of 842 ms. Nevertheless, the region merging process, which lowered the segment count from 17 to 11, significantly reduced unnecessary transitions and enhanced spatial continuity, thus justifying the slightly elevated computational expense.

In comparison, Map-5, despite containing a central obstacle and areas with mixed orientations, recorded the shortest total path length $L_{total} = 1237$, the lowest intra-coverage cost $L_\alpha = 69$, and the quickest execution time (316 ms). These findings indicate that compact segment distributions with well-aligned connectivity paths result in better performance in terms of coverage efficiency and real-time functionality.

Table 3. Performance metrics for different test maps.

Map	# Seg-ments	L_a	L_{int}	L_{total}	% Coverage	Duration (ms)
Map-1	(17 $\rightarrow$ 11)	797	33,333	34,130	98.6	842
Map-2	11	137	2428	2565	96.8	412
Map-3	11	236	2030	2266	97.3	389
Map-4	22	575	1514	2089	96.1	561
Map-5	7	69	1168	1237	98.1	316

Map-2 and Map-4, which feature repetitive and elongated structures, show an increase in intra-segment costs due to their narrow and stretched segment shapes. However, their reduced inter-region costs and segment regularity lead to acceptable coverage rates (96.8% and 96.1%, respectively) within moderate time frames.

Map-3 provides a balanced structure that is particularly useful for evaluating inter-region transition planning. With an obstacle centrally positioned that necessitates a symmetric transition design, the algorithm maintained a solid balance between path length and responsiveness.

In summary, the findings imply that the proposed framework is especially effective in semi-structured environments where obstacles are either sparse or consistently arranged, and the segment geometry facilitates strategic merging or alignment. The adaptive inter-region planning component guarantees that transitions are performed dynamically, resulting in coverage ratios that consistently exceed 96% across all scenarios.

All time measurements for the various algorithms were conducted in a MATLAB R2024a environment on a system equipped with an Intel Core i5 (11th generation) processor, 32 GB of DDR4 RAM, and an Intel Arc graphics card (Intel Corp., Santa Clara, CA, USA). During the assessment, the algorithm was executed solely on the CPU with no use of GPU acceleration. On this hardware setup, it was observed that the proposed method finished in an average of 842 ms for scenarios with high transition costs, such as Map-1, and in 316 ms for simpler structures like Map-5. These results indicate the computational viability of the method for real-time applications.

The suggested region-based segmentation combined with Boustrophedon coverage and Obstacle-Aware Inter-Region Transition-based transition planning was benchmarked against two prominent basic approaches from the literature. For performance evaluation, a comparative analysis was conducted against the Global Boustrophedon [12,22,39,41] and Spiral Decomposition (Spiral CPP) [40] approaches, which have been presented in the literature as robust and efficient algorithms for coverage path planning:

- Global Boustrophedon Path Planning (G-BPP): A classical technique that examines the obstacle-free area as a unified whole.
- Cell-by-Cell Spiral Decomposition (Spiral CPP): A pattern that traverses all map cells in a spiral fashion and alters direction in localized areas.

Map-3 was selected for this comparison due to its structure, which is neither overly complex nor overly simplistic in terms of the number of segments and placements of obstacles. It features both narrow passages and uniform segment orientations, presenting mixed environmental characteristics that can evaluate the proposed method's performance both within segments and between them. It is understood that spherical or spiral methodologies in the literature generally do not perform optimally in such geometries. The assessments for each method were conducted on the same system configuration under identical starting and ending conditions. The durations reported only account for the path planning process, excluding visualization or I/O operations.

Table 4 displays the results of comparative evaluations of the proposed approach alongside two established baseline methods—Global Boustrophedon Path Planning (G-BPP) and Spiral Coverage Path Planning (Spiral CPP)—on Map-3, characterized by multiple obstacle zones containing narrow corridors and complex free-space areas.

Table 4. Comparative performance outputs of the methods on Map-3 (highlighting strengths of the proposed method).

Method	Segment Structure	L_a	L_{int}	L_{total}	% Coverage	Duration (ms)
Proposed Method	Partial	236	2030	2266	97.3	389
Global Boustrophedon (G-BPP) [36]	Single part	1986	-	1980	87.5	645
Spiral CPP	Single part	2145	-	2145	92.1	568

The Proposed Method utilizes a hybrid strategy for region segmentation that combines rectangular decomposition with partial merging. It achieves a notably reduced intra-region coverage length $L_a = 236$ and inter-region connection cost $L_{int} = 203$, culminating in a total path length $L_{total} = 2266$ that is significantly shorter than those reported by the global spiral and boustrophedon methods, which have total lengths of 2145 and 1980, respectively.

While G-BPP shows a slightly lower overall distance than the proposed method, its coverage ratio (87.5%) is significantly lower, indicating substantial unmapped areas, likely due to interruptions caused by obstacles in the global sweeping pattern. Spiral CPP achieves better coverage (92.1%) but suffers from a longer path length due to its redundant looping characteristics. In contrast, the proposed method secures a high coverage rate of 97.3%, showcasing superior adaptability to the topology of the map while minimizing unnecessary overlaps. Furthermore, the computational time of 389 ms demonstrates the method's efficiency, as it is considerably quicker than G-BPP (645 ms) and Spiral CPP (568 ms), emphasizing the scalability and real-time applicability of the proposed technique. These results indicate that a hierarchical segmentation strategy combined with inter-region optimization not only improves area coverage but also offers more compact and efficient paths, particularly in complicated indoor or semi-structured settings. The findings support the need for integrating inter-region coordination strategies—such as partial merging and local CPP path planning—as essential elements in advanced coverage frameworks.

To further interpret the outcomes presented in Table 4, each performance metric is analyzed in terms of its contribution to the overall system efficiency as follows:

- **Coverage:** In the proposed method, the ratio of unskipped areas remains high due to segment-based coverage. The Global Boustrophedon and Spiral methods tend to omit certain regions in narrow passages characterized by complex obstacle arrangements.
- **Total Cost (L_{total}):** The transitions in the suggested approach are optimized using the Obstacle-Aware Inter-Region Transition algorithm, resulting in a more balanced sum of the internal and external paths. In contrast, the Spiral method has a longer inner coverage, but its overall cost may be lower due to the absence of transitions.
- **Duration:** The proposed method achieves a time advantage as its path planning is more localized, attributed to its segment-based structure.

The five experimental maps (Map-1 through Map-5) were intentionally chosen to represent a wide array of coverage scenarios commonly encountered in indoor robotic applications. Map-1 illustrates a large-scale, sparse layout that highlights lengthy paths within segments and costly transitions between regions. Maps 2 and 4 model structured grid environments characterized by repetitive, elongated segments—common in warehouse shelving or crop row inspection, making them suitable for evaluating narrow coverage strips and maintaining region continuity. Map-3 features a symmetrical hybrid design

focused around an obstacle, providing a standard for assessing balanced transitions and symmetric path planning. Lastly, Map-5 showcases a semi-structured layout with varying geometric orientations and a significant central obstruction, resembling realistic mixed-use spaces such as dynamic offices or adaptable industrial areas. These configurations were selected to comprehensively test the proposed strategies for region growing, merging, and obstacle-aware transitions across different levels of complexity and layout irregularities.

5.2. Comparative Evaluation with Standard A* Planner

To further assess the effectiveness and practical use of the proposed segment-based coverage planning algorithm, comparative analysis was performed against the traditional A* planner on a realistically scaled map (Map-1 see Figure 4), measuring 30 m × 30 m and divided into a 268 × 268 grid. This environment was structured to simulate an industrial warehouse setting, featuring wide corridors and sparse partitioned areas—making it an ideal test environment for evaluating both local coverage efficiency and global transition behavior.

Table 5 provides a comprehensive comparison across seven essential performance metrics: intra-segment path length (L_α), total path length (L_{total}), path overlap ratio (POR), execution time (Duration), number of direction changes, memory utilization, and scalability on larger grid maps. The results reveal several significant differences:

Table 5. Performance metrics comparing the proposed region-guided planner and the A*-based baseline on Map-3, highlighting improvements in path efficiency, execution time, and resource usage.

Metric	Proposed Method	A*	Unit
L_α (Intra-segment)	1418	2445	grid
L_{total} (Total Path)	1646	2445	grid
POR	2.7%	18.1%	%
Duration	486	705	ms
Dir. Changes	42	111	count
Mem. Usage	9.7	17.3	MB
Scalability (100 × 100)	1.92	3.66	sec

Efficiency in Coverage: The suggested method achieves a notably reduced intra-segment path length ($L_\alpha = 1418$) in comparison to the conventional A* algorithm ($L_\alpha = 2445$), attributable to its organized Boustrophedon sweeps across clearly defined rectangular areas.

Global Optimality: Despite the total path length (L_{total}) accounting for both coverage and inter-region transitions, the proposed technique still yields a lower total (1646 vs. 2445), showcasing its efficient coordination across segments.

Minimal Redundancy: The overlap ratio of the path (POR) is significantly decreased from 18.1% (A*) to 2.7%, demonstrating that the proposed method minimizes redundant revisits to areas that have already been covered.

Execution Time and Resources: This new approach demands roughly 31% less processing time (486 ms vs. 705 ms) and utilizes less memory (9.7 MB vs. 17.3 MB), emphasizing its computational efficacy.

Trajectory Smoothness: The count of directional changes is substantially lower (42 vs. 111), suggesting smoother navigation paths that could lead to reduced actuator strain and lower energy use.

Scalability: When applied to a larger grid (100 × 100), the proposed algorithm exhibits improved scalability, taking nearly half the time compared to the standard A* (1.92 s vs. 3.66 s).

As illustrated in Table 5, these findings collectively highlight the strength and efficiency of the proposed framework in structured and semi-structured settings, particularly where reducing path redundancy, computational duration, and mechanical effort are essential operational objectives.

5.3. Complexity Analysis of the Algorithm

The structure of the proposed algorithm consists of three key components:

- Rectangular segmentation through Region Growing,
- Row-wise Boustrophedon coverage path generation for each segment,
- Solution for transition paths between segment centers utilizing the proposed algorithm.

Each component's computational complexity is individually analyzed. These analyses are supported by both theoretical asymptotic complexity and empirical timing results.

Region Segmentation (Region Growing):

- Input: $n \times n$ dimensional grid map
- Process: Rectangular expansion into surrounding empty space by visiting each cell no more than once
- Complexity: $\mathcal{O}(n^2)$

As each cell is processed only once, and expansion operations are executed with constant-time linear checks. In combined segment configurations like Map-1, the additional "region merging" procedure does not increase complexity since it involves linear scanning and area overlap checks on segment sets.

Boustrophedon Coverage Path Generation:

A zig-zag scan (Boustrophedon) is conducted row by row for each rectangular segment.

- For an average segment size of $h_i \times w_i$: $\mathcal{O}(h_i \cdot w_i)$ per segment.
- Total for all segments: $\sum_{i=1}^{N_r} \mathcal{O}(h_i w_i) = \mathcal{O}(n^2)$.

This is because the overall processing load for covering all free space is constrained by the total number of cells in the map.

Transition Between Segments:

For total N_r segments:

- The nearest connection point is determined each step.
- Transition complexity: $\mathcal{O}(n \log n)$ (assuming it operates on a grid in an obstacle-rich environment)
- For transition sequence total: $\mathcal{O}(N_r \cdot n \log n)$

Nevertheless, in the experimental scenario, since Obstacle-Aware Inter-Region Transition is only applied between segment centers and is restricted to valid cells, the actual complexity is: $\mathcal{O}(N_r \cdot d \log d)$, ($d \ll n$)

Total Complexity of the Algorithm

The combined complexity for the entire algorithm flow is: $\mathcal{O}(n^2 + N_r \cdot d \log d)$

Here:

- n^2 : Maximum burden from grid size
- N_r : Number of segments ($n \ll N_r \ll n^2$)
- d : Average number of valid cells between segments (relatively small in sparse grids)

This structure allows the algorithm to maintain an acceptable computational load for real-time applications, particularly for medium to large grid sizes (50×50 – 200×200).

6. Discussions

In prior stages of the coverage algorithm, it is presumed that the Boustrophedon path created for each rectangular area concludes at both ends of the area (starting and ending points). These terminal points of the path are identified as connection nodes (segment terminals) within the global coverage path. When the end segments from all areas are eliminated in this manner, the challenge that emerges is to link these segment ends to form the global coverage route. As part of the suggested method, a directional Boustrophedon path is generated for each rectangular coverage area; this path moves from the starting point to cover every free cell individually, with turns based on rows. In the example depicted in Figure 9, the paths generated across four distinct regions are illustrated, and the order of transition between regions is established according to the solution of the Traveling Salesman Problem (TSP). Each sequence of transitions ($1 \rightarrow 2 \rightarrow 3$) is represented by dashed lines. Furthermore, obstacles (shown as gray areas) are represented as restrictive zones that constrain both the coverage direction within regions and the path for transitions between regions. Consequently, the global coverage path is designed in an integrated fashion considering both local optimizations and transition costs.

Problem Definition: Given N rectangular regions, each represented by the following segment (Equation (16)):

$$S_i = (p_i^{\text{start}}, p_i^{\text{end}}), i = 1, 2, \dots, N \quad (16)$$

Here:

- p_i^{start} and $p_i^{\text{end}} \in R^2$: denotes the initial and final coordinates of the intra-regional path.
- Total length for each segment: $L(S_i) = |p_i^{\text{start}} - p_i^{\text{end}}|$

The objective is to sequence the connection paths between these segment ends, factoring in the obstacles present on the map while minimizing the overall transition cost.

Cost Model: A valid path γ_{ij} is determined between the endpoints of two segments employing the A* algorithm, taking obstacle details into account, with its cost calculated as follows (Equation (17)):

$$C_{ij} = \text{Cost}(S_i \rightarrow S_j) = |\gamma_{ij}| + L(S_j) \quad (17)$$

Here:

- $|\gamma_{ij}|$: Length of the passageway identified by A*
- $L(S_j)$: Internal path length of the newly incorporated segment

The total system cost can be represented as follows (Equation (18)):

$$\min_{\pi \in \mathcal{P}} \left(\sum_{k=1}^N L(S_{\pi_k}) + \sum_{k=1}^{N-1} |\gamma_{\pi_k, \pi_{k+1}}| \right) \quad (18)$$

Here:

- π : A permutation that indicates the order of connecting segments
- $\mathcal{P}$: A collection of all valid segment arrangements

The start and ending points of the Boustrophedon path formulated within each rectangular coverage area are identified as the segment ends of that area. These endpoints act as nodes for the formation of the global coverage route. To minimize the overall transition costs between segments, a connectivity analysis is executed among these nodes and valid routes are generated while considering obstacles. Although traditional full optimization

methods based on the Traveling Salesman Problem (TSP) offer theoretically optimal solutions for such connectivity challenges, they quickly escalate the computational complexity, particularly as the number of segments increases. Thus, the traditional A* assisted method offers a greedy selection approach prioritizing computational efficiency along with the obstacle-awareness, yielding a practical and modular solution. The algorithm begins by fixing the starting segment and assessing the transition paths from the exit end of the current segment to the endpoints of the unused segments utilizing the A* algorithm. The total cost for each transition is computed by taking into account both the A* path length and the coverage path length of the newly added segment. The lowest cost among these options is selected, and the next segment is incorporated into the trajectory. Segments can be connected in various orientations (from start to finish or the reverse); the algorithm automatically selects the more advantageous orientation. This iterative process continues until all segments are covered. This arrangement can be interpreted as a directed, weighted graph problem, where the nodes represent segment endpoints and the edges indicate valid transition paths discovered through the A* algorithm.

While the conventional A*-assisted segment transition strategy offers a robust framework regarding computational efficiency and usability, it does have certain limitations. These limitations become especially noticeable in environments with intricate obstacle arrangements and tight passage scenarios:

- **Decision-Making Based on Geometric Distance:** Even though the A* algorithm enhances passage routes between segment endpoints while accounting for obstacles, the overall route planning is predominantly influenced by geometric distances. This can result in a preference for routes that are shorter in distance but have a high density of obstacles.
- **Non-Directional Heuristic Strategy:** The heuristic values used in the passage decisions do not consider the robot's current heading, which can lead to unnecessary turns and zigzag maneuvers for robots that are less maneuverable. This can increase both energy usage and the total time required for the task.
- **Possibility of Getting Trapped in Temporary Optima:** A greedy selection method might favor the least expensive local passage at each decision point, even if it results in more expensive choices later on. This raises the likelihood of straying from the global optimum.
- **Neglect of Obstacle Density:** Although the A* algorithm avoids traversing obstacles, it does not incorporate the environmental impact of obstacle density (such as narrow pathways within a constricted corridor) into its cost function. This may result in risky or confined areas being overlooked.
- **Scalability Issues:** In extensive maps with an increasing number of segments, the computation of each passageway using A* incurs a greater computational cost. This cost can become significant, particularly during the selection of directions and the assessment of passage alternatives.

For these reasons, a more advanced passage cost model has been developed to enhance both passage stability and minimize total coverage time, except in simpler maps where the existing system is effective. In this regard, a multi-weighted passage cost function is introduced in the second version of the study, which takes into account environmental density of obstacles, changes in orientation, and geometric distance simultaneously. To enhance the understanding of the performance trade-offs associated with the proposed framework, Table 6 outlines the primary benefits and drawbacks of its core components, which encompass region segmentation, intra/inter-region planning, and computational expenses. This table provides a clear perspective on the method's advantages and areas where improvements can be made.

Table 6. Summary of Advantages and Limitations of the Proposed Framework.

Aspect	Advantage	Limitation
Map Decomposition [7,10,12,13]	Region Growing facilitates adaptive rectangular partitioning, enhancing the suitability for boustrophedon approaches.	It may generate numerous small regions in cluttered settings if merging is not applied.
Intra-Region Coverage [31]	Boustrophedon traversal ensures comprehensive coverage with minimal changes in direction.	It assumes the presence of static and flat surfaces; variations in slope or height have not been accounted for yet.
Inter-Region Planning [35]	Utilizes a multi-cost TSP approach that considers directional bias and hurdle proximity.	The TSP-derived route may not be ideal in highly dynamic or chaotic settings.
Region Merging (FSM) [13,14,22]	Combines less effective smaller areas based on fill-efficiency, minimizing coverage fragmentation.	Adjustments to the threshold setting (τ) may be necessary based on map density and dimensions.
Computational Cost [36,38]	Demonstrates efficiency (e.g., an average runtime of 486 ms) and is scalable for real-time applications.	Currently, it is implemented in MATLAB; deploying in real-time might require adapting to embedded systems.
Modularity [7,8,28]	Each phase (segmentation, CPP, transitions) can be upgraded separately.	Dependencies between modules might create integration challenges in customized platforms.
Comparison with Baselines [7,38,40]	Surpasses Spiral and A*-based CPP in terms of path length, coverage, and execution speed.	The benchmarking is confined to structured and semi-structured indoor environments.

Integration with Real-Time Trajectory Planning: Although the framework presented emphasizes high-level coverage decomposition and route optimization, its modular structure allows for incorporation with low-level real-time trajectory planners. After creating the global coverage path (which includes intra-region Boustrophedon paths and inter-region TSP transitions) either offline or in a semi-online manner, a reactive trajectory controller (like Model Predictive Control (MPC), the Dynamic Window Approach (DWA), or Time Elastic Band (TEB)) can be utilized to follow the path in dynamic settings. Additionally, the FSM module can be enhanced to observe real-time data such as obstacle proximity, localization drift, or battery status, and initiate local replanning for specific segments while maintaining coherence in the overall route. This ensures that the proposed approach is suitable for real-time operation on field robots with limited onboard processing capabilities.

7. Conclusions

In this research, a modular and flexible framework for coverage path planning based on regions was introduced, featuring automatic rectangular segmentation, Boustrophedon-based coverage for individual areas, and an obstacle-aware transition planner that is directionally biased for navigating between regions. Unlike traditional methods that depend solely on Euclidean distance or standard A* search algorithms, the proposed transition planner combines geometric distance, obstacle density, and heading changes into a comprehensive multi-weighted cost model. This facilitates more realistic and efficient connections between segments, especially in cluttered or narrow environments.

The approach was tested on five different grid maps that depict structured indoor settings, elongated agricultural fields, and mixed scenarios. The results of the experiments showed consistently high coverage rates, ranging from 96% to 99%, with overall planning times between 316 and 842 ms on a typical hardware system. Implementing region

merging on complex maps (e.g., Map-1) considerably minimizes the number of transitions, enhancing both efficiency and trajectory compactness.

A comparative evaluation on Map-3 indicates that the proposed framework surpasses traditional global coverage methods and spiral scanning techniques, particularly in cases of spatial fragmentation. The advantages of the algorithm are found in its sensitivity to obstacles, directionally informed transition choices, and its appropriateness for real-time robotic applications.

Despite its strengths, there are certain limitations—such as the restriction to axis-aligned rectangular regions, dependence on heuristic ordering of regions, and occasional performance decline when there are too many segments. Nevertheless, the suggested framework provides a practical and adaptable solution for coverage planning in both structured and semi-structured settings, establishing a foundation for future developments involving adaptive segmentation, multi-robot coordination, and terrain-aware motion planning.

8. Future Work

To enhance the applicability and robustness of the proposed coverage planning framework based on regions, several important research avenues will be explored:

Generalization to Freeform Segmentation: Future iterations of the zoning module will offer the ability to create convex and concave polygonal sub-regions. This improvement is designed to facilitate effective coverage in areas with irregular shapes, curved edges, or obstacles that are not aligned with the axes.

Dynamic environments: While the proposed framework shows significant efficiency and modularity in static, structured settings, its current design presumes that obstacle arrangements stay constant throughout the mission. This presumption is valid in various real-world environments, like warehouses, greenhouses, or semi-structured indoor spaces, where changes in layout are minimal. However, in scenarios where obstacles can emerge or shift unexpectedly (such as the presence of humans or dynamic storage systems), this constraint may hinder real-time adaptability. To address this challenge, the framework could incorporate a reactive replanning mechanism, enabling the inter-region transition module to utilize incremental planning algorithms like D* Lite or Anytime D* to adjust to environmental changes. These enhancements can be seamlessly integrated into the modular architecture without modifying the region segmentation or intra-region coverage components, maintaining the overall design while improving adaptability.

Slope- and Energy-Aware Coverage: The plan is to incorporate terrain slope and the energy consumption of robots into both the coverage and movement planning processes. This will enable the planner to develop routes that are more energy-efficient, particularly in outdoor environments or on uneven ground.

Global Region Ordering via Metaheuristics: Instead of depending on greedy methods, the sequencing of global segments will be refined using meta-heuristic techniques such as Genetic Algorithms (GA), Simulated Annealing (SA), or Ant Colony Optimization (ACO). The goal is to reduce the total costs of coverage and transitions through a globally informed strategy.

Extension to Multi-Robot Coordination: The proposed framework will be modified for multi-robot systems, tackling challenges like cooperative allocation of sub-regions, collision avoidance, and dynamic reassignment of tasks in both uniform and diverse robotic teams.

Real-World Implementation and Testing: The algorithm will be executed on a tangible mobile robot that has capabilities for localization, mapping, and avoiding obstacles. Experimental trials will be carried out to evaluate the robustness of the system in real-world scenarios that include sensor inaccuracies, actuator errors, and partial observation of the environment.

Author Contributions: Conceptualization, S.K.; Methodology, S.K.; Software, S.K. and M.Z.K.; Validation, S.K. and M.Z.K.; Formal analysis, S.K.; Investigation, S.K.; Resources, S.K. and M.Z.K.; Data curation, S.K. and M.Z.K.; Writing—original draft, S.K. and M.Z.K.; Writing—review and editing, S.K. and M.Z.K.; Visualization, S.K.; Supervision, M.Z.K.; Project administration, M.Z.K.; Funding acquisition, M.Z.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research did not receive any external funding.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Acknowledgments: This study was conducted at the MAVI Laboratory of the Faculty of Engineering. The authors confirm that there was no external funding obtained for this study.

Conflicts of Interest: The authors declare no conflict of interest.

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